Md. Aman Uddin Siyam

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Summary

Graduated with a BSc in Computer Science & Engineering from IUBAT, equipped with practical experience building machine learning and deep learning solutions for computer vision, biomedical signal processing, and time-series tasks. Skilled in the end-to-end development lifecycle—from data collection and preprocessing to model training, evaluation, and interpretability using SHAP and Grad-CAM. Experienced in deploying models via containerized services and REST APIs using Docker and FastAPI/Flask. Proficient in Python, TensorFlow/Keras, NumPy, and Pandas, with working knowledge of PyTorch and experiment tracking. Seeking an **Internship** in **Machine Learning Engineering** or **MLOps** in Dhaka to contribute research-driven ML features and production-ready pipelines.

Key Skills

Programming	Python, NumPy, Pandas, Matplotlib, scikit-learn.
Deep Learning & CV	TensorFlow/Keras (proficient), PyTorch (familiar), OpenCV, transfer learning, Grad-CAM.
Signal & Time-series	Biomedical signal processing, feature extraction, ARIMA, Prophet, LSTM, PCA, survival analysis.
MLOps & Deployment	Reproducible pipelines, experiment logging, Docker (basic), FastAPI/Flask (familiar), REST API integration, W&B (learning).
Data & Research	Exploratory data analysis (EDA), feature engineering, cross-validation, evaluation metrics, experimental design, SHAP explainability.
Tools & Platforms	Google Colab, Kaggle, Git / GitHub, Jupyter Notebooks.
Soft skills	Team leadership, communication, mentoring, time management, collaboration.
Languages	Bangla (native), English (fluent).

SELECTED PROJECTS

Financial RAG Analyst

Engineered a production-grade RAG pipeline to analyze SEC 10-K filings using **LlamaIndex**, **Qdrant**, and **Llama 3**. Architected an **AI Observability** system with a secure dashboard to monitor hallucinations and capture **RLHF** user feedback. Integrated **LlamaParse** for complex table extraction, containerized via **Docker**, and implemented **GitHub Actions CI/CD** pipelines.
GitHub: Financial-RAG-Analyst | Live App

BAT-Track: Energy-Efficient Visual Tracking for UAVs

Developed a lightweight computer vision pipeline for real-time object tracking on resource-constrained UAVs. implemented state-of-the-art trackers (**OSTrack**, **SiamAPN++**) optimized for edge deployment, significantly reducing computational latency and power consumption while maintaining high tracking accuracy during rapid aerial maneuvers.
GitHub: BAT-Track

Edge-ViT-FSL: Resource-Aware Few-Shot Learning for Cross-Domain Plant Disease Diagnosis on IoT Devices

CPU-optimized, cross-domain crop disease diagnosis system using a lightweight MobileViT backbone and saliency-guided few-shot learning to identify plant pathologies in "wild" environments with minimal training

data.
GitHub: Edge-vit-disease-detect | Live App

Automated PCOS Detection from Ultrasound

Engineered a Deep Learning pipeline to detect PCOS in ultrasound imaging automatically. Utilized Transfer Learning with Convolutional Neural Networks .
GitHub: Ultrasound-PCOS

Real Estate Management System

Built a full-stack MERN Real Estate Management System with Buyer/Seller/Admin roles and an admin approval pipeline for listings. Implemented property CRUD, map integration for location-based viewing, favorites, inquiries/reviews modules, and an email notification system. Deployed the React (Vite) frontend on Vercel and the Node/Express API on Render with MongoDB persistence.
GitHub: Real-Estate-Management-System | Live App

Experience

Co-Founder & Researcher	CollabCircle: Lead collaborative ML/DL research projects, coordinate experiments, mentor contributors, and manage roadmaps.

Research & Publications (selected)

- Voice-Based Parkinson's Disease Detection using SVM & XGBoost. (Paper Link)
- Clinical vs. Ultrasound for PCOS Detection: A Calibrated, Leakage-Safe, Real-Time Benchmark (Accepted at ICCIT).
- Reproducible Machine Learning Pipeline for Predicting Cosmetic Chemical Carcinogenicity and Reproductive Toxicity Using Molecular Descriptors and Fingerprints (Accepted at ICCIT).

CERTIFICATIONS

- **Machine Learning with Python - EDGE Bangladesh (Nov 2024 - May 2025)** Certificate
- **Kaggle Courses:** Computer Vision, Intro to Deep Learning, Intermediate ML, Intro to Machine Learning,Pandas,Data Visualization, Python (2025) Certificate
- **Introduction to generative AI (Google Cloud, 2025)** Certificate
- **AI for Beginners - HP Life (2025)** Certificate

Training & Workshops

- Machine Learning with Python - EDGE Bangladesh (Nov 2024 - May 2025) Certificate

Awards & Activities

- Participant - Solvio AI Hackathon (2025)
- Participant - ICPC Dhaka Regional Preliminary Contest (2024)

EDUCATION

2022-present	BSc in Computer Science & Engineering, **IUBAT**	(CGPA: 3.24 current)
2020	HSC, **UHSC**	(GPA: 4.67)
2018	SSC, Bright Sun High School	(GPA: 4.72)

Hobbies

- Playing cricket
- Fishing

Last updated: January 26, 2026